

Initial Value Problems

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$$F(x, y, y', y'', \dots, y^{(n)}) = 0 \leftarrow n\text{-th order ODE}$$

ODE

Nonlinear ODEs

Linear ODEs

$$a_0(x) y^{(n)} + a_1(x) y^{(n-1)} + \dots + a_{n-1}(x) y' + a_n(x) y = F(x)$$

Homogeneous
ODEs
($F(x) = 0$)

Non-homogeneous
ODEs
($F(x) \neq 0$)

$$y'' + p(x)y' + q(x)y = 0$$

linear homogeneous
ODE of 2nd
order.

$$y'' + y = F(x)$$

$$y_1, y_2 \in \text{sol}^n$$

$$y_1 + y_2 \notin \text{sol}^n$$

$$F(x) = x^2$$

Q7.

Is $y=0$ a solⁿ of

$$y'' + y = x^2 \text{ in } [1, 2] = I$$

Ans - No

$$y'' + y = 0$$

$$y_1, y_2 \in \text{sol}^n$$

$$y_1 + y_2 \in \text{sol}^n$$

$$y_1 + cy_2 \in \text{sol}^n, \forall c \in \mathbb{R}$$

$$y=0 \in \text{sol}^n$$

Seperable ODE

$$y' = f(x, y) \stackrel{?}{=} h(x) g(y)$$

$$\Rightarrow \int \frac{dy}{g(y)} = \int h(x) dx$$

$$\frac{x^2 y^2}{1+x^2}$$

$$y' = \frac{dy}{dx} = \frac{x^2 y^2}{1+x^2}$$

$$= \underbrace{\left(\frac{x^2}{1+x^2} \right)}_{h(x)} \underbrace{y^2}_{g(y)}$$

Initial Value Problem (IVP)

- The distinct requirements of a practical problem involve more than merely a differential equation. We also need to specify suitable auxiliary conditions that define the problem being studied. Specifically, we are interested in the scenario of the initial value problem defined for an n -th order differential equation as outlined below.

Definition (Initial Value Problem)

An n -th order ODE together with n -auxiliary conditions of the form

$$y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1},$$

where $y_0, y_1, \dots, y_{n-1}$ are constants, is called an initial value problem.

$$a_0(x) y^{(n)} + a_1(x) y^{(n-1)} + \dots + a_1 y = F(x)$$

$$y'' + y = 0.$$

$$\Rightarrow y = y(x) = c_1 \sin x + c_2 \cos x.$$

2nd order

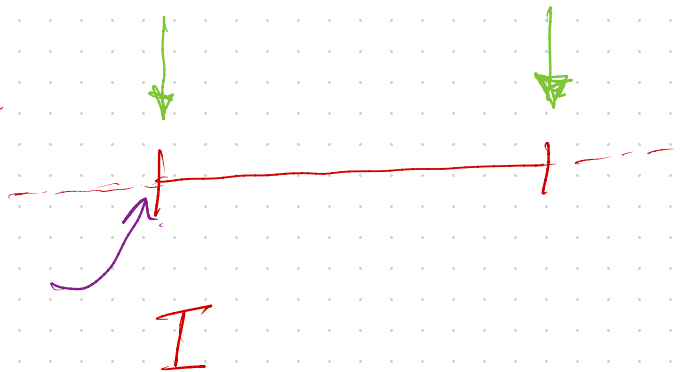
$$y(a) = \alpha$$

$$y'(a) = \beta$$

$$\begin{array}{c} | \quad | \\ 0 \quad 1 \end{array}$$

$$y(0) = \alpha$$

$$y'(0) = \beta.$$



Example of IVP

Example

Find the solution to the initial value problem

$$y'' = e^{-x}, \quad y(0) = 1, \quad y'(0) = 4.$$

$$y'' = e^{-x}$$

$$y = e^{-x} + \boxed{c}x + \boxed{d}$$

$$y(0) = 1$$

$$y'(0) = 4$$

for any values
of c & d
 y will be a solⁿ
of $y'' = e^{-x}$; hence
a general solⁿ

$$y(x) = e^{-x} + 5x + 0$$

Normal Form

- IVPs play an important role in the theory and applications of ODEs. The previous ODE has a unique solution. More generally, suppose we have a differential equation that can be written in the normal form

$$y^{(n)} = f(y', y'', \dots, y^{(n-1)}).$$

- According to the definition (1), the IVP for such an n -th order ODE is the following: Solve

$$y^{(n)} = \cancel{f(y', y'', \dots, y^{(n-1)})} \quad f(x, y, y', \dots, y^{(n-1)})$$

subject to

$$y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1},$$

where $y_0, y_1, \dots, y_{n-1}$ are constants.

ODE: $F(x, y, y', y'') = 0$

normal
form.

$$y'' = f(x, y, y')$$

$$y' = f(x, y)$$

— this is also a
normal form

$$y'' + y^2 - \sin xy = 0$$

— order 2

$$F(x, y, y', y'') = y'' + y^2 - \sin(xy)$$

$$\Rightarrow F(x, y, y', y'') = 0$$

— Given ODE

$$\begin{aligned} \Rightarrow y'' &= \sin(xy) - y^2 \\ &= f(x, y, y') \end{aligned}$$

$$f(x, y, y') = \sin(xy) - y^2$$

Geometry of First-Order ODEs

- The primary aim of this chapter is to study the first-order differential equation

$$\frac{dy}{dx} = f(x, y), \quad (1)$$

where $f(x, y)$ is a given function of x and y .

- Our focus is on the geometric aspects of the differential equation and its solutions.

Definition (Solution Curve)

The graph of any solution to the differential equation (1) is called a solution curve.

Geometry of First-Order ODEs

Remark

- If we recall the geometric interpretation of the derivative $\frac{dy}{dx}$ as giving the slope of the tangent line at any point on the curve with equation $y = y(x)$, we see that the function $f(x, y)$ in (1) gives the slope of the tangent line to the solution curve passing through the point (x, y) .
- Consequently, when we solve (1), we are finding all curves whose slope at the point (x, y) is given by the function $f(x, y)$.
- According to our definition, the general solution to the differential equation (1) involves one arbitrary constant; geometrically, it gives a family of solution curves in the xy -plane, one for each value of the arbitrary constant.

$$\frac{dy}{dx} = y' = f(x, y)$$

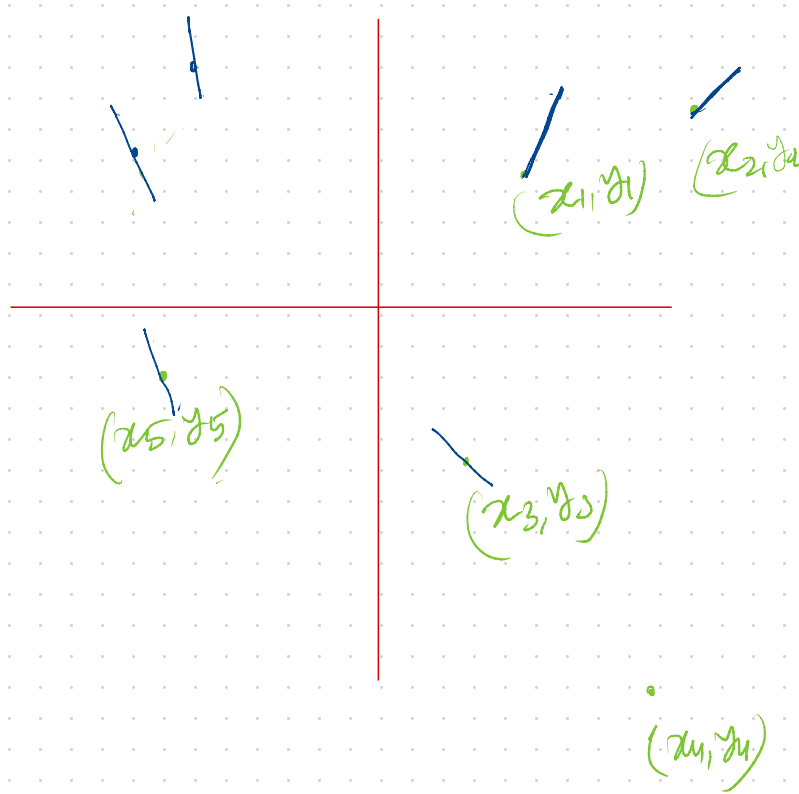
$y \rightarrow$ unknown

$f \rightarrow$ already provided

$$\left. \frac{dy}{dx} \right|_{(x_1, y_1)} = f(x_1, y_1) \quad (=100)$$

$$\left. \frac{dy}{dx} \right|_{(x_2, y_2)} = f(x_2, y_2) \quad (=1)$$

$$\left. \frac{dy}{dx} \right|_{(x_3, y_3)} = f(x_3, y_3) \quad (-4)$$



Solution Curve

Example

Find the general solution to the differential equation $\frac{dy}{dx} = 2x$, and sketch the corresponding solution curves.

Direction Fields

$$\left. \frac{dy}{dx} \right|_{(1,1)} = 2$$

$$\left. \frac{dy}{dx} \right|_{(2,1)} = 4$$

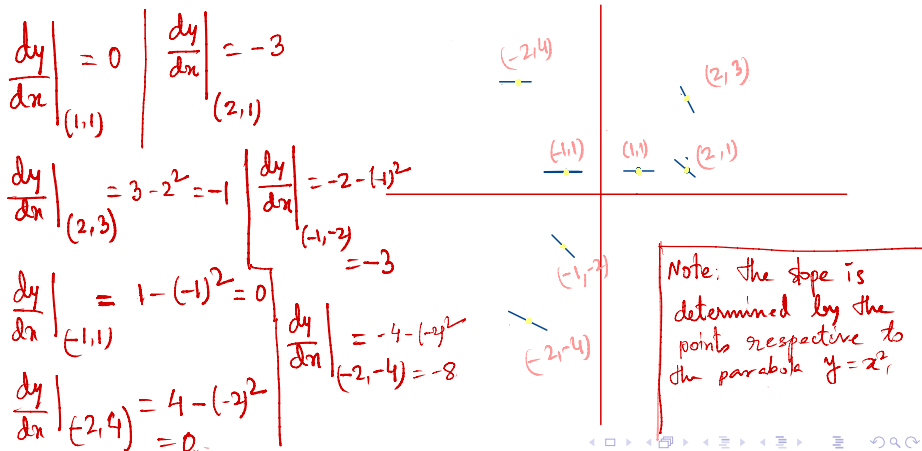
$$\left. \frac{dy}{dx} \right|_{(-2,1)}$$



Solution Curve

Example

Find the general solution to the differential equation $\frac{dy}{dx} = y - x^2$, and sketch the corresponding solution curves.



Existence & Uniqueness

- It is useful for the further analysis of the differential equation (1) to give at this point a brief discussion of the *existence and uniqueness* of solutions to the corresponding initial-value problem

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0. \quad (2)$$

- Geometrically, we are interested in finding the particular solution curve to the differential equation that passes through the point in the xy -plane with coordinates (x_0, y_0) . The following questions arise regarding the initial-value problem:

- 1 **Existence:** Does the initial-value problem have any solutions?
- 2 **Uniqueness:** If the answer to question 1 is yes, does the initial-value problem have only one solution?

Example:- Consider the IVP

$$\frac{dy}{dx} = y^{1/3} \quad \text{and} \quad y(0) = 0.$$

Let us define two functions:

$$y_1(x) = f_1(x) = 0 \quad \forall x$$
$$\text{and } y_2(x) = f_2(x) = \begin{cases} \left(\frac{2}{3}x\right)^{3/2}, & x \geq 0 \\ 0 & x < 0, \end{cases} \quad \left| \begin{array}{l} y_2'(x) = \quad, x < 0 \\ y_2'(0) = \quad \\ y_2'(x) = \quad, x > 0. \end{array} \right.$$

are both solⁿs of the initial value problem. There could be infinitely many solⁿs for the same problem.

$$R \ y_2'(0) = \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{2^{3/2} h^{3/2}}{h} = 2^{3/2} \lim_{h \rightarrow 0^+} h^{1/2} = 0.$$

$$L \ y_2'(0) = \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{0 - 0}{h} = 0.$$

$$f(x, y) = y^{1/3}$$

— cont^s.

$$\frac{\partial f}{\partial x} = 0 \quad \left| \quad \frac{\partial f}{\partial y} = \frac{1}{3} y^{-2/3} \right.$$

$$f(x, y)$$

$$f_x(x, y), f_y(x, y)$$

$$f_{xx}(x, y), f_{xy} = f_{yx}, f_{yy}$$

$$\frac{\partial f}{\partial x}(x, y) = 0 = g(x, y)$$

$$\frac{\partial f}{\partial y}(x, y) = \frac{1}{3} y^{-2/3} = h(x, y)$$

f is cont^s $\Rightarrow \frac{1}{f}$ is also cont^s
whenever $f(x) \neq 0$

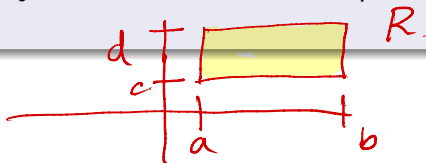
$\frac{\partial f}{\partial y}$ is not cont^s on $(x, 0)$
— x-axis

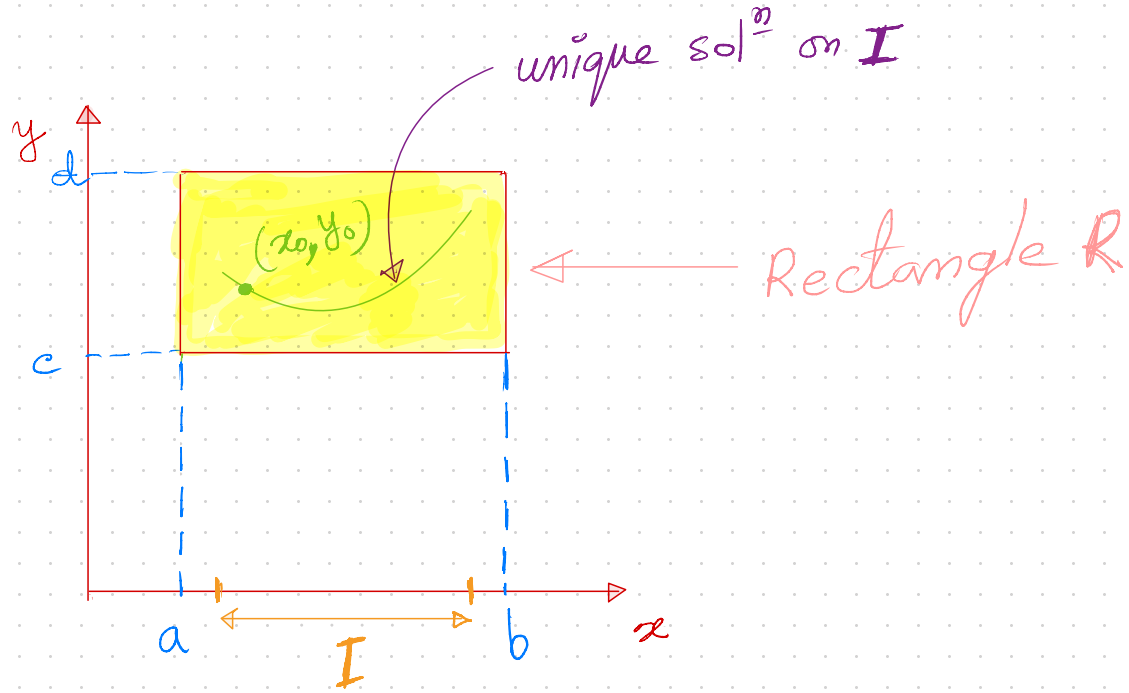
Existence & Uniqueness

- Certainly, in the case of an applied problem, we would be interested only in the IVPs that have precisely one solution. The following theorem establishes conditions on f that guarantee the existence and uniqueness of a solution to the initial-value problem

Theorem (Existence and Uniqueness Theorem)

Let $f(x, y)$ be a function that is continuous on the rectangle $R := \{(x, y) : a \leq x \leq b, c \leq y \leq d\}$. Suppose further that $\frac{\partial f}{\partial y}$ is continuous in R . Then for any interior point (x_0, y_0) in the rectangle R , there exists an interval I containing x_0 such that the initial-value problem (2) has a unique solution for x in I .





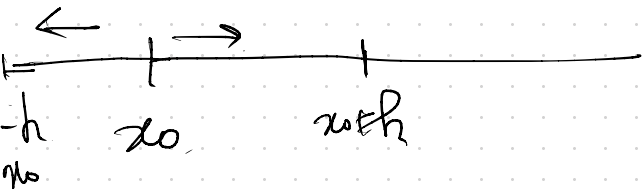
Let R be a rectangle defined as $-a \leq x - x_0 \leq a$
 $\Rightarrow |x - x_0| \leq a$ and $|y - y_0| \leq b$.

Let $f \in C(R)$. Then the IVP
 $y' = f(x, y), \quad y(x_0) = y_0$

has a solⁿ $y = y(x)$. This solⁿ is defined for all
 x in the interval $|x - x_0| \leq h$, where

$$h = \min \left\{ a, \frac{b}{M} \right\}, \quad \text{where } M = \max_{(x,y) \in R} |f(x,y)|$$

$$= \sup_{(x,y) \in R} |f(x,y)|$$

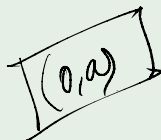


Example: Existence & Uniqueness Theorem

Example

Prove that the initial-value problem

$$\frac{dy}{dx} = 3xy^{1/3}, \quad y(0) = a$$

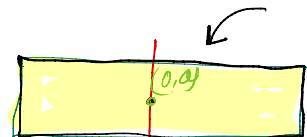


has a unique solution whenever $a \neq 0$.

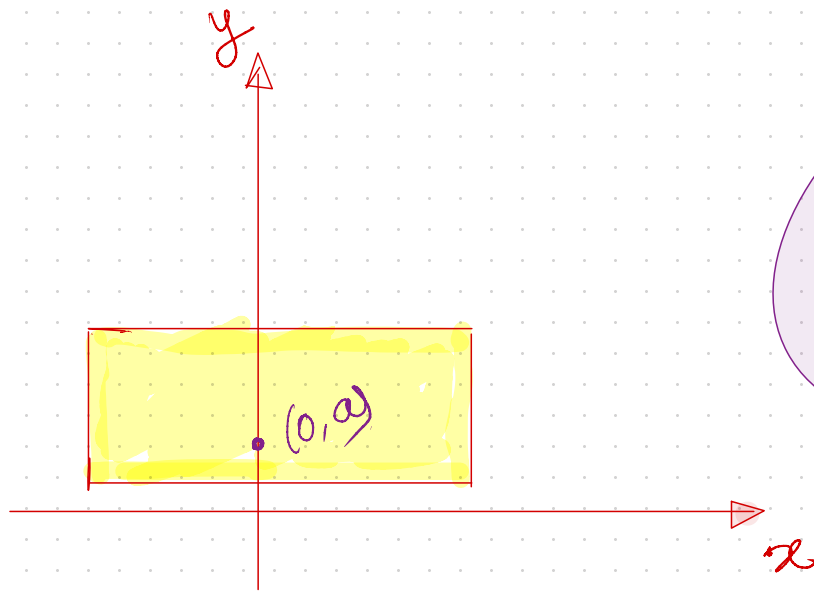
Solⁿ. $f(x, y) = 3xy^{1/3}$; then f is cont^s in $\mathbb{R}^2$.
Therefore, there is at least one solⁿ.

$$\text{Now } \frac{\partial f}{\partial y} = 3 \cdot \frac{1}{3} x y^{-2/3}$$

Since $\frac{\partial f}{\partial y}$ is cont^s in the region R containing $(0, a)$,



$(0, 0)$ we can find such a region not containing $(0, 0)$



$a \neq 0$

the yellow shaded rectangle will imply the unique solⁿ

But if we place a bigger rectangle which intersects the x -axis, we will be in trouble.

⊙ Remark: The conditions of existence and uniqueness theorem are sufficient but not necessary. For example, we consider $y' = \sqrt{y} + 1$, $y(0) = 0$, $x \in [0, 1]$

(Next slide)

⊙ Example: Consider the IVP $\left. \begin{array}{l} y' = x^2 + y^2 \\ y(1) = 3 \end{array} \right\}$

Now $f(x, y) = x^2 + y^2$. Then $\frac{\partial f}{\partial y} = 2y$. Both the f 's f and $\frac{\partial f}{\partial y}$ are cont^2 in every domain D of the xy -plane. The initial condⁿ $y(1) = 3$ means $x_0 = 1$ and $y_0 = 3$, and the point $(1, 3)$ certainly lies in some domain D . Hence all hypotheses are satisfied $\Rightarrow$ unique solⁿ exists on some interval $|x - 1| \leq h$ about $x_0 = 1$, which satisfies the initial condⁿ.

Example:- solⁿ.

$$y' = \sqrt{y} + 1$$

$$y(0) = 0$$

$$\Rightarrow \left[\int \frac{dy}{\sqrt{y}+1} = \int dx. \right]$$
$$y = v^2$$

Define $z(x) = (\sqrt{y_1(x)} - \sqrt{y_2(x)})^2$

By diff, we get

$$z'(x) = -2(\sqrt{y_1(x)} - \sqrt{y_2(x)}) \left(\frac{1}{2\sqrt{y_1(x)}} y_1'(x) - \frac{1}{2\sqrt{y_2(x)}} y_2'(x) \right)$$

$$= -(\sqrt{y_1} - \sqrt{y_2}) \left(\frac{\sqrt{y_1} + 1}{\sqrt{y_1}} - \frac{\sqrt{y_2} + 1}{\sqrt{y_2}} \right)$$

$$= -\frac{(\sqrt{y_1} - \sqrt{y_2})^2}{\sqrt{y_1 y_2}} \Rightarrow \boxed{z'(x) \leq 0}$$

$$z(x) \geq 0, \quad z'(x) \leq 0, \quad z(0) = 0$$

$\boxed{z = 0}$

Existence

$f(x, y)$ is cont^s for $(x, y) \in \mathbb{R}^2$
with $y \geq 0$



$\frac{\partial f}{\partial y}$ is not
cont^s $\Rightarrow$ no com.
on uniqueness

— existence is
fine!

Vague Idea

P-2. Theorem

$$\left. \begin{array}{l} y' = f(x, y) \\ y(x_0) = y_0 \end{array} \right\}$$

$\Leftrightarrow$

$$y(x) = \int_{x_0}^x f(t, y(t)) dt + y_0$$

diff eqnⁿ

Integral eqnⁿ

$$\{y_0(x), y_1(x), y_2(x), \dots\}$$

$\updownarrow$
 y

$\updownarrow$
 y

$\updownarrow$
 y

$$y_1(x) = \int_{x_0}^x f(t, y_0(t)) dt + y_0$$

$$y_2(x) = \int_{x_0}^x f(t, y_1(t)) dt + y_0$$

Example:- Let $R = \{(x, y) : |x-0| \leq 3, |y-0| \leq 3\}$ be a rectangle.

Let $y' = f(x, y)$ where $f(x, y) = xy$. Then $f \in C(R)$
 $y(0) = 0$.

Then, the IVP has at least one solⁿ. The solⁿ is defined for all x in the interval $|x-0| \leq h$, where $h = \min\{3, 3/K\}$; $K = \max_{(x,y) \in R} |xy|$.

Since $f(x, y) = xy \Rightarrow \frac{\partial f}{\partial x} = y$ and $\frac{\partial f}{\partial y} = x$.

So extremum in the interior is at $x=0$ & $y=0$, i.e., at $(0,0)$. The max^m can occur at $x=3, x=-3, y=3, y=-3$ lines of the region. Hence max^m will occur at $(\pm 3, \pm 3)$ and the value is $K=9$. $h = \min\{3, 3/9\} = 1/3$.

$$f(x, y) = xy$$

$$\{(x, y) \in \mathbb{R}^2 : |x| \leq 3, |y| \leq 3\}$$

R

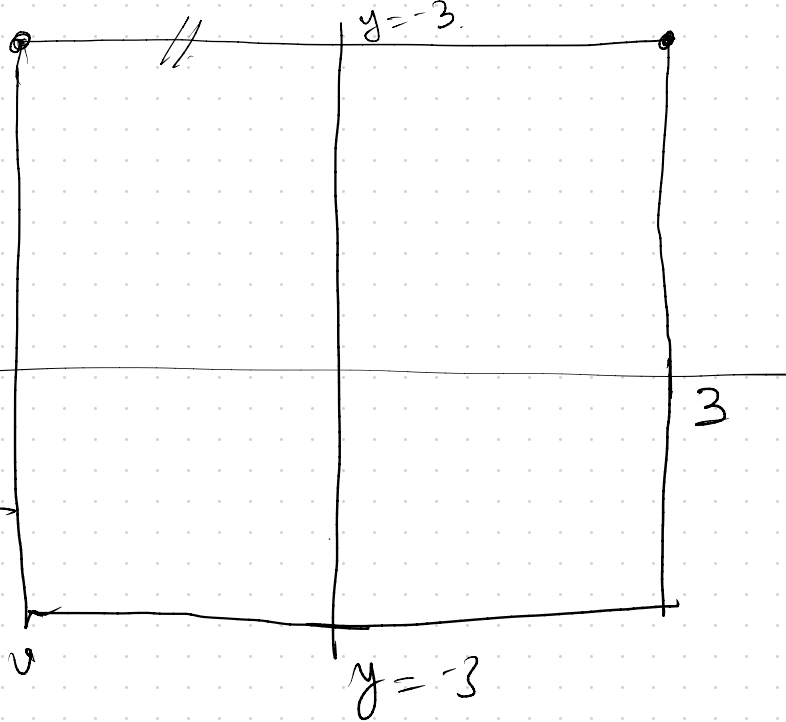
for $\max^m / \min^m$ at interior of $R = \{(x, y) \in \mathbb{R}^2 : |x| < 3, |y| < 3\}$

$$f_x = y \text{ and } f_y = x$$

$$\text{For } \max^m / \min^m \quad f_x = 0 \text{ and } f_y = 0$$

$$\Rightarrow x = 0 \text{ and } y = 0 \quad \left[\text{2nd derivative test for extremals} \right]$$

$$\text{For } \boxed{y = 3} \quad f(x, 3) = 3x$$



Thank You

Questions?